

```

//FIFO

#include <stdio.h>

int main() {

    int n, frames;

    // =====
    // INPUT NUMBER OF PAGES
    // =====

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    /*
    Example:

    n = 7 pages

    Reference String:
    1 3 0 3 5 6 3
    */

    // =====
    // DECLARE REFERENCE STRING ARRAY
    // =====

    int ref[n];

    // =====
    // INPUT REFERENCE STRING
    // =====

    printf("Enter the reference string: \n");

    for(int i = 0; i < n; i++) {
        scanf("%d", &ref[i]);
    }

    // =====
    // INPUT NUMBER OF FRAMES
    // =====

```

```
printf("Enter number of frames: \n");
scanf("%d", &frames);
```

```
/*
```

Frames = Number of memory slots available

Example:

```
frames = 3
```

```
*/
```

```
// =====
```

```
// DECLARE FRAME ARRAY
```

```
// =====
```

```
int frame[frames];
```

```
// Initialize frames as empty (-1)
```

```
for(int i = 0; i < frames; i++) {
```

```
    frame[i] = -1;
```

```
}
```

```
// =====
```

```
// FIFO POINTERS AND COUNTERS
```

```
// =====
```

```
int front = 0; // points to the oldest page in frames
```

```
int faults = 0; // counts page faults
```

```
int hits = 0; // counts page hits
```

```
// =====
```

```
// FIFO PAGE REPLACEMENT LOOP
```

```
// =====
```

```
for(int i = 0; i < n; i++) {
```

```
    int found = 0; // flag to check if page is already in frame
```

```
    // =====
```

```
    // CHECK FOR PAGE HIT
```

```
    // =====
```

```
    for(int j = 0; j < frames; j++) {
```

```
        if(frame[j] == ref[i]) {
```

```

    /*
    Page is already in memory
    → HIT
    */

    found = 1;
    hits++;
    break;
}
}

// =====
// PAGE FAULT HANDLING
// =====

if(!found) {

    /*
    Page is not in memory → FAULT

    FIFO Replacement:

    - Replace the oldest page (pointed by `front`)
    - Move `front` to next frame cyclically
    */

    frame[front] = ref[i];
    front = (front + 1) % frames;

    faults++;
}

}

// =====
// FINAL OUTPUT
// =====

printf("\nTotal Page Faults: %d\n", faults);
printf("Total Page Hits: %d\n", hits);

return 0;
}

```

LRU:

```
#include <stdio.h>
```

```
int main() {
    int n, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    int pages[n];
    printf("Enter the reference string: \n");
    for(int i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: \n");
    scanf("%d", &frames);

    int frame[frames], time[frames];

    // Initialize
    for(int i = 0; i < frames; i++) {
        frame[i] = -1;
        time[i] = 0;
    }

    int pageFaults = 0, pageHits = 0, counter = 0;

    for(int i = 0; i < n; i++) {
        int found = 0;

        // Check for Hit
        for(int j = 0; j < frames; j++) {
            if(frame[j] == pages[i]) {
                found = 1;
                pageHits++;
                counter++;
                time[j] = counter; // update recent use
                break;
            }
        }
    }
}
```

```

// Page Fault
if(!found) {
    int lruIndex = 0;

    // Find least recently used
    for(int j = 1; j < frames; j++) {
        if(time[j] < time[lruIndex]) {
            lruIndex = j;
        }
    }

    counter++;
    frame[lruIndex] = pages[i];
    time[lruIndex] = counter;
    pageFaults++;
}

printf("\nTotal Page Faults: %d\n", pageFaults);
printf("Total Page Hits: %d\n", pageHits);

return 0;
}

```